5. It has high resistance to heat and can be made as insulating material.
6. It has low thermal conductivity.
7. It can be moulded into various shapes.
8. It also sticks well to fibrous surfaces.
9. It does not interact chemically with paint and prevents alkali attack.
10. Its gypsum content provides smoothness and shine.

Disadvantages of Plaster of Paris

The following are the disadvantages of using Plaster of Paris:

1. Gypsum plaster is not suitable for exterior finish as it is slightly soluble in water.
2. It is more expensive than cement or cement-lime plaster.
3. It cannot be used in moist situations.
4. Skilled labour is required for precise application and thus labour cost for applying plaster of Paris is high.

2.8.5 Sound Insulating materials

Noise due to traffic, aircraft, barking dogs., equipment, machinery, etc., affects our peaceful and quiet living atmosphere. These divert our concentration resulting in stress, fatigue and less productivity. To overcome these problems, some type of sound insulating materials have to be used in the construction.

On the basis of acoustic properties, materials can be divided in to the following groups:

1. Sound reflection materials
2. Sound absorbing materials
3. Sound insulation materials

1. Sound reflection materials

These materials are generally placed in the ceiling, back of the stage and side walls of lecture halls. Materials such as wood, special plaster and concrete are sound reflection materials. The sound reflection property of materials is expressed by their sound reflection coefficients.

2. Sound absorbing materials

Absorption is expressed by the absorption coefficient. The materials with loose structure such as carpets, wool mats, perforated hard boards (where the sound energy is lost in the holes), etc., are called typical sound absorbing material.

3. Sound insulation materials

Dense materials such as bricks do not allow sound to pass through. There are many types of construction such as solid brick walls, cavity walls, double wall partitions with sound absorbing materials in between. The sound insulation property of a material is expressed by the reduction of noise in decibel.

2.8.6 Electrical insulating materials

The material which does not allow the electricity to pass through them is known as electrical insulating material. It provides the high resistive path to the electric current through which it is impossible to pass through. It is also used in the overhead transmission line.

The commonly used electrical insulating materials are glass, plastic, rubber and wood.

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